TSAI CHENGYU (Patrick)

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EDUCATION

NATIONAL TAIWAN UNIVERSITY

Information Management, College of Management | Sep 2024 - Expected Jun 2028

- GPA: Fall 2025 4.20/4.3; Spring 2026 4.14/4.3. Sophomore-year rank: 7/49.
- Relevant coursework: HCI and User Research (A+), Web Application Development (A+), Statistical Learning and Deep Learning (A+), Statistics I & II (A+), Management (A).

SELECTED PROJECTS

AI MISTAKE LEARNING ASSISTANT PLATFORM

Project Lead, System Analysis & Integration | Spring 2026

- Led requirements interviews with tutoring-center staff; translated workflows into functional and data requirements.
- Planned system architecture, data models and REST APIs; coordinated integration across student, teacher and Gemini AI components.
- The team delivered a functional prototype for AI Q&A, mistake records, follow-up practice and learning analytics.

STATISTICAL LEARNING PROJECTS

Model Implementation & Analysis | Fall 2025

- Combined TF-IDF, structured features, class weights and threshold search for job-talent classification using Logistic Regression and LightGBM. Reported validation F1 improved from approximately 0.29 to 0.57; final LightGBM F1: 0.5696.
- Used text normalization, character n-grams, HashingVectorizer and Ridge Regression for product-price prediction; evaluated with five-fold cross-validation.

MULTIMODAL TRANSIT PLANNING - HCI RESEARCH

User Research & Analysis Team Member | Spring 2026

- Contributed to interviews and analysis in a team study with 11 participants, using semi-structured interviews, think-aloud testing and thematic analysis.
- Synthesized route-comparison and transfer pain points into interface recommendations; presented the work in English.

MULTI-AGENT HITL SUPPORT SYSTEM

CEO-side Integration, Deployment Support & Simulation | 2026

- Helped connect Telegram, Google Apps Script and Gmail for a simulated support workflow with department and CEO approval. This was a course prototype using fictional company data.

LEADERSHIP & EXPERIENCE

NTU INFORMATION MANAGEMENT STUDENT ASSOCIATION

Finance Director | Sep 2025 - Jun 2026

- Managed budgets and spending; coordinated departmental resource needs to support activity planning within budget constraints.

Additional experience: Planning team, NTU College of Management Student Association (2025-2026); NTU IM Camp PR & administration, including insurance and coach-service vendor communication (2026).

SKILLS

Systems & AI: Requirements interviews, REST API design, data modeling, Gemini API integration, Google Apps Script, Telegram/Gmail integration.

Data & research: Python, pandas, text features, scikit-learn, LightGBM, Ridge, OLS, model evaluation; semi-structured interviews, think-aloud testing and thematic analysis.